CHAPTER 20

NATURAL HAZARDS AND DISASTERS

CHAPTER ESSENCE AT A GLANCE:
Examine hazard versus disaster concepts, seismic risk zones of India (Zones II to V), tsunami wave shoaling mechanisms, recurring flood causes in Indo-Gangetic basins, meteorological and agricultural droughts, and the Disaster Management Act 2005.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Hazard versus Disaster Distinction:** A natural hazard is an extreme physical event with potential to cause damage; it transforms into a disaster only when it impacts a vulnerable human population lacking capacity to absorb the impact without external aid.

• **Seismic Zoning Architecture of India:** Nearly 59 percent of India's landmass is vulnerable to moderate to severe earthquakes; categorized by Bureau of Indian Standards into 4 Seismic Zones: Zone II (low), Zone III (moderate), Zone IV (severe), and Zone V (very severe like Himalayas and Kutch).

• **Floods, Droughts and Hydrological Extremes:** Floods inundate over 40 million hectares annually (Brahmaputra and Ganga basins); Droughts progress sequentially from meteorological (rain deficit) to hydrological (water body drying) to agricultural drought (soil moisture stress).

• **Modern Disaster Management Paradigm:** Shifted under the Disaster Management Act 2005 from post-disaster relief to proactive pre-disaster mitigation, structural engineering codes, community capacity building, and early warning networks under NDMA and SDMAs.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Disaster Risk Equation} = \frac{\text{Hazard Event} \times \text{Socioeconomic Vulnerability}}{\text{Community Coping Capacity}}$$

Fundamental Quantitative Disaster Risk Formulation

$$\log_{10}(E) = 4.8 + 1.5M \rightarrow \text{Earthquake Energy Multiplier } (\approx 31.6\times \text{ per Unit Magnitude})$$

Gutenberg-Richter Logarithmic Seismic Energy Scaling

$$v_{\text{tsunami}} = \sqrt{g \cdot h} \rightarrow \text{Deep Ocean Wave Velocity } (\approx 700 \text{ to } 800 \text{ km hr}^{-1})$$

Lagrangian Shallow-Water Tsunami Wave Propagation Velocity

$$\text{Disaster Continuum: Prevention and Mitigation} \rightarrow \text{Preparedness} \rightarrow \text{Response} \rightarrow \text{Recovery}$$

Four-Phase Integrated Disaster Management Continuum

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** India does not have a Seismic Zone I; the Bureau of Indian Standards revised and merged Zone I into Zone II, so the active seismic risk zones are Zone II, Zone III, Zone IV, and Zone V.

• **Alert:** Tsunamis in the deep open ocean have small wave amplitudes (often under 1 metre) and are virtually undetectable by ships; they rear up into towering destructive walls of water only when entering shallow coastal waters due to shoaling.

• **Alert:** A meteorological drought (lack of rainfall) leads to a hydrological drought (drying up of rivers, lakes, and reservoirs), which subsequently triggers an agricultural drought (soil moisture deficiency causing crop failure).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between a Natural Hazard and a Natural Disaster. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Natural Hazard:** An extreme natural physical event or condition (like a cyclone, earthquake, or flood) that possesses the potential to cause injury, damage to property, or environmental disruption. **[1 Mark]**

• **Natural Disaster:** The actual catastrophic occurrence when a hazard strikes an unprepared, vulnerable human settlement, causing widespread loss of life, livelihood, and property that exceeds local coping capacity. **[1 Mark]**

2. Mention the four Seismic Zones of India as classified by the Bureau of Indian Standards (BIS). Which zone represents the highest risk? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Four Seismic Zones:** Zone II (Low damage risk), Zone III (Moderate damage risk), Zone IV (High damage risk), and Zone V (Very high damage risk). **[1 Mark]**

• **Highest Risk Zone:** Zone V represents the highest seismicity and destruction potential (covers Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Rann of Kutch, North-East India, and Andaman and Nicobar Islands). **[1 Mark]**

3. Explain the three types of droughts recognized in India (Meteorological, Hydrological, and Agricultural Drought). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Meteorological Drought:** A situation where there is a significant prolonged deficiency of precipitation over an area compared to its normal long-term average. **[1 Mark]**

• **Hydrological Drought:** Occurs when prolonged meteorological drought causes water levels in storage reservoirs, lakes, streams, and groundwater tables to dry up below acceptable thresholds. **[1 Mark]**

• **Agricultural Drought:** Occurs when extreme soil moisture deficiency results in wilting of crops, severe yield reduction, and complete agricultural harvest failure. **[1 Mark]**

4. What are the primary natural and human causes of recurring devastating floods in the Brahmaputra and Ganga river basins? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Torrential Monsoon Downpours:** Excessive, highly concentrated rainfall during the southwest monsoon saturates the upper Himalayan catchments within a short period. **[1 Mark]**

• **Severe Siltation and Shallow Beds:** Heavy deforestation in steep Himalayan catchments causes massive soil erosion, choking river channels with silt and drastically reducing their carrying capacity. **[1 Mark]**

• **Floodplain Encroachment:** Unplanned human encroachment onto natural floodplains, construction of ill-designed embankments, and destruction of natural wetlands that act as flood cushions. **[1 Mark]**

5. Describe the mechanism, propagation velocity, and coastal impact of Tsunamis with special reference to the 2004 Indian Ocean Tsunami. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Mechanism:** Giant seismic sea waves generated by sudden vertical displacement of the seafloor during underwater megathrust earthquakes, marine volcanic eruptions, or submarine landslides. **[1 Mark]**

• **Deep Ocean Propagation:** In deep oceans, tsunamis travel at jet airliner speeds (700 to 800 km per hour) with tiny wave heights (< 1 metre) and enormous wavelengths exceeding 200 km. **[1 Mark]**

• **Shoaling Effect on Coast:** As waves enter shallow coastal waters, wave speed drops drastically while wavelength compresses, causing wave height to rear up into towering destructive walls of water 15 to 30 m high. **[1 Mark]**

• **2004 Indian Ocean Tsunami:** Triggered on 26 December 2004 by a 9.1 magnitude earthquake off northern Sumatra; devastated coastal Tamil Nadu (Nagapattinam), Kerala (Kollam), and Andaman and Nicobar Islands. **[1 Mark]**

6. Discuss the key stages of the Disaster Management Cycle and the institutional framework established under the Disaster Management Act 2005 in India. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Disaster Management Cycle:** Encompasses four continuous phases: (1) Mitigation (structural and policy risk reduction), (2) Preparedness (early warning, drills, stockpiles), (3) Response (immediate search, rescue, evacuation, and medical relief), and (4) Recovery (rehabilitation and building back better). **[2 Marks]**

• **National Disaster Management Authority (NDMA):** Apex statutory body headed by the Prime Minister to lay down policies and guidelines for disaster management. **[1 Mark]**

• **State and District Authorities:** State Disaster Management Authorities (SDMAs headed by Chief Ministers), District Disaster Management Authorities (DDMAs headed by District Collectors), and the National Disaster Response Force (NDRF) for specialized emergency deployment. **[1 Mark]**